

ICT 2233 - Web Application Development

Assignment 02

Deadline All components must be Completed **On or before 24th September 2025**

Submission Instructions

1. **GitHub Classroom:** Accept the assignment invitation via the provided link. All work must be committed to your designated classroom repository.
<https://classroom.github.com/a/kK1yCvkF>
2. **Deploy :** Website must be live via GitHub Pages.
3. **Submit :** Provide your GitHub repo link + live site URL.
4. **AI Usage Rule:**
 - AI tools may be used for **debugging and syntax help**.
 - Overuse of AI-generated code will result in **mark deduction**.

Objective

Build a personal portfolio website using React.js. You will start from a React template and fully customize it with your own content, a design system, and modern features.

This assignment is designed to test your proficiency in several key areas of modern web development:

- **Modern CSS Styling:** Flexbox, Grid, responsive design, Tailwind/Sass, and animations.
- **JavaScript Fundamentals:** DOM manipulation, events, async **fetch** API, and control flow.
- **React Concepts:** Components, props, state, hooks, and routing.
- **Git Practices:** Branch management, meaningful commits, and collaboration standards.
- **Deployment:** Hosting on GitHub Pages with a professional README.

By the end of this assignment, you'll have a professionally deployed personal portfolio to showcase your skills to potential employers.

Requirements

1. Portfolio Content & Structure

- **Home Page:** Must include a hero section with your introduction, name, tagline, and a professional photo.
- **About Page:** Detail your background, education, and both technical and soft skills.
- **Projects Page:** Showcase at least 3 projects, each with:
 - A screenshot or preview image.
 - A short, compelling description.
 - A link to the GitHub repository or a live demo.
- **Contact Page:** A working contact form with validation and links to your LinkedIn, GitHub, and email.
- **404 Page:** A custom error page with a link to navigate back to the Home page.

2. Advanced CSS & Styling

- Apply the **Box Model**, control **specificity**, and use **semantic selectors**.
- Utilize **Flexbox & CSS Grid** for creating robust layouts.
- Ensure a **mobile-first responsive design** using media queries, `clamp()`, and `minmax()`.
- Apply at least **two** modern styling tools from the following:
 - **Tailwind CSS** (utility-first classes).
 - **Sass/SCSS** (nesting, mixins, variables).
 - **CSS Modules** or **Styled Components** for scoped styling.
- Implement a cohesive **design system** (colors, typography, spacing, buttons).
- Add a **dark mode toggle** using React state and CSS variables/Tailwind.
- Use **Framer Motion** or **CSS keyframes** for smooth animations and page transitions.

3. JavaScript Features

- Demonstrate proficiency with variables, functions, loops, and control flow.
- Implement **event handling** (e.g., dark mode toggle, form validation, interactive buttons).
- Use the **fetch API** with **async/await** to pull live data from an external source (e.g., display your GitHub repositories dynamically).
- Show an understanding of the **event loop** and asynchronous behavior.

4. React.js Features

- Build the entire application using **functional components**.
- Pass information between components via **props**.
- Manage component state with the **useState** hook.
- Handle side effects, such as API calls, with the **useEffect** hook.
- Implement client-side navigation using **React Router**.
- **(Bonus)**: Create and use a **custom hook**.

5. GitHub & Deployment

- Initialize the project with Git from the start.
- Follow strict **Git best practices**:
 - Commit frequently with clear, meaningful messages.
 - Maintain a clean and professional commit history.
- Host the final project on **GitHub Pages**.
- Create a professional **README.md** file that includes:
 - A project description.
 - Installation and usage instructions.
 - A list of technologies used.
 - The live demo link from GitHub Pages.

6. File Structure (Sample)

```
portfolio-react/
├── public/
│   └── index.html
├── src/
│   ├── assets/           # Images, icons, fonts
│   ├── components/       # Reusable UI parts (Navbar, Footer, ProjectCard)
│   ├── pages/            # Main page components (Home, About, Projects, Contact)
│   ├── styles/           # Tailwind, Sass, or CSS Modules
│   └── App.js            # Main application component with routing
├── package.json
├── README.md
└── .gitignore
```

~Think in components. Build with passion. Go for it!~